Jonathan Lys

Research / representations / generation

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Third-year PhD candidate at IMT Atlantique. Research on discrete diffusion, diversity-aware decoding, and EEG foundation models.

Experience

Visiting Scientist / 2024 - 2025

Sony Europe, Stuttgart, Germany

Two visits to EUREC, total of 3 months.

- Worked on conditional generation for text diffusion models.
- Studied looped AR LLMs for reasoning tasks.

Research Intern / 2024-04 - 2024-10

Criteo AI Lab, Paris, France

Studied diffusion models, for image background generation.
Supervised by Jaspreet Sandhu, Thibaut Issenhuth, Ludovic Dos Santos

- Trained large-scale diffusion models up to 6B parameters.
- Developed an ad background generation pipeline, reducing object extension by 10×, achieving lowest FID and fast inference.
- Worked on billions of product images, leveraging PySpark and distributed training.

Research Intern / 2023

IMT Atlantique, BRAIN Team, Brest, France

Studied PEFT methods for SAM and attention-based data augmentation techniques.
Supervised by Bastien Pasdeloup.

- Matched the performance of full fine-tuning with a custom LoRA method.
- Improved the performance of CNNs through attention-based data augmentation.

Developer Intern / 2022

French Polynesia's government IT Department, Papeete, French Polynesia

Development of a Web application for network management, automation of database migrations.

Education

PhD in Computer Science / 2024 - 2027

IMT Atlantique - BRAIn team, Brest

- Studying discrete diffusion models for text generation and diversity-aware sampling strategies.
- Co-authored a NeurIPS paper on foundation model for EEG and worked on its release.
- Served as a reviewer for ICML (Silver Reviewer Award).

M.Sc. Engineering Sciences and Applications / 2023 - 2024

University of Rennes 1, Rennes

- Double Degree SIVOS in Engineering, jointly accredited by Rennes 1 and CentraleSupélec.
- Specializing in Signal, Image, and Vision. Graduated with high honours.

Engineering Degree / 2021 - 2024

IMT Atlantique, Brest

- Specializations: Digital Transformation and Innovation, then Mathematical and Computational Engineering.
- Coursework in machine learning, stochastic modeling, numerical methods, deep learning, data science, and computer science.

Postgraduate Study Abroad / 2023-02 - 2023-06

University of Technology Sydney, Australia

- Academic exchange: Deep learning and convolutional neural networks, computational intelligence, IT projects management, decision-making for engineers.

Publications

TensLoRA: Tensor Alternatives for Low-Rank Adaptation (2026)

Marmoret, Axel, Bensaid, Reda, Lys, Jonathan, Gripon, Vincent, Leduc-Primeau, François

Stacked LoRA for Subject-Adaptive EEG Foundation Models in Motor Imagery Decoding (2026)

Sarhane, Aymen, Lbakali, Fouad, Souissi, Mouad, Lys, Jonathan, Lioi, Giulia

Key Ingredients for EEG Foundation Models: What Works - and What Doesn't (2026)

Lioi, Giulia, El Ouahidi, Yassine, Lys, Jonathan, Thölke, Philipp, Pasdeloup, Bastien, Gripon, Vincent, Farrugia, Nicolas, Jerbi, Karim

How Transferable Are EEG Foundation Models? A Case Study on Sleep Staging (2026)

Lamouroux, Alix, Benoit, Lucas, Lys, Jonathan, Farrugia, Nicolas, Lioi, Giulia

Inner Loop Inference for Pretrained Transformers: Unlocking Latent Capabilities Without Training (2026)

Lys, Jonathan, Gripon, Vincent, Pasdeloup, Bastien, Marmoret, Axel, Mauch, Lukas, Cardinaux, Fabien, Hacene, Ghouthi Boukli

Residual Connections and the Causal Shift: Uncovering a Structural Misalignment in Transformers (2026)

Lys, Jonathan, Gripon, Vincent, Pasdeloup, Bastien, Marmoret, Axel, Mauch, Lukas, Cardinaux, Fabien, Hacene, Ghouthi Boukli

D5P4: Partition Determinantal Point Process for Diversity in Parallel Discrete Diffusion Decoding (2026)

Lys, Jonathan, Gripon, Vincent, Pasdeloup, Bastien, Marmoret, Axel, Mauch, Lukas, Cardinaux, Fabien, Hacene, Ghouthi Boukli

REVE: A Foundation Model for EEG - Adapting to Any Setup with Large-Scale Pretraining on 25,000 Subjects (2025)

El Ouahidi, Yassine, Lys, Jonathan, Thölke, Philipp, Farrugia, Nicolas, Pasdeloup, Bastien, Gripon, Vincent, Jerbi, Karim, Lioi, Giulia

FICUS: Few-shot Image Classification with Unsupervised Segmentation (2024)

Lys, Jonathan, Lin, Frédéric, Béliveau, Clément, Decaestecker, Jules, Bendou, Yassir, Abdali, Aymane, Pasdeloup, Bastien

Skills

Programming: Python, SQL, PySpark, C++, Julia, R, Java, Matlab, Bash, Linux shell

Deep Learning: PyTorch, HuggingFace, TensorFlow, Keras, Expertise: CNN, ViT, DiT, MAE, LLM, dLMM

Data & ML: NumPy, Pandas, Scikit-learn, Statistical & Probabilistic modeling, Scientific watch

Distributed Systems: Compute: SLURM, Torch Distributed, HF Accelerate, Data: Apache Spark, Apache Hadoop

Tools & Dev: Git, GitHub/GitLab (CI/CD workflows), Jupyter, Office Suite, LaTeX

Soft Skills: Applied Research, Agile Methods, Decision-making, Technical Communication

Extra-curricular Activities

IMTA Research Committee - PhD Student Representative

IMTA PhD Student Association - Secretary

IMTA Photography Association - President (until 2023)

IMTA Short Film Association - President (until 2023)

IMTA Sound & Light Association - Officer (until 2023)

IMTA Arts Association - Member (until 2023)

IMTA Chess Association - Member

Languages

English: C1, IELTS 7.5

Mandarin: B2.1

Spanish and German: Basic

French: Native

Interests

Music: violin for 12 years

Competitive chess: World top 0.3% online

Photography: Fujifilm X-S10 + XF 35mm f/1.4 R